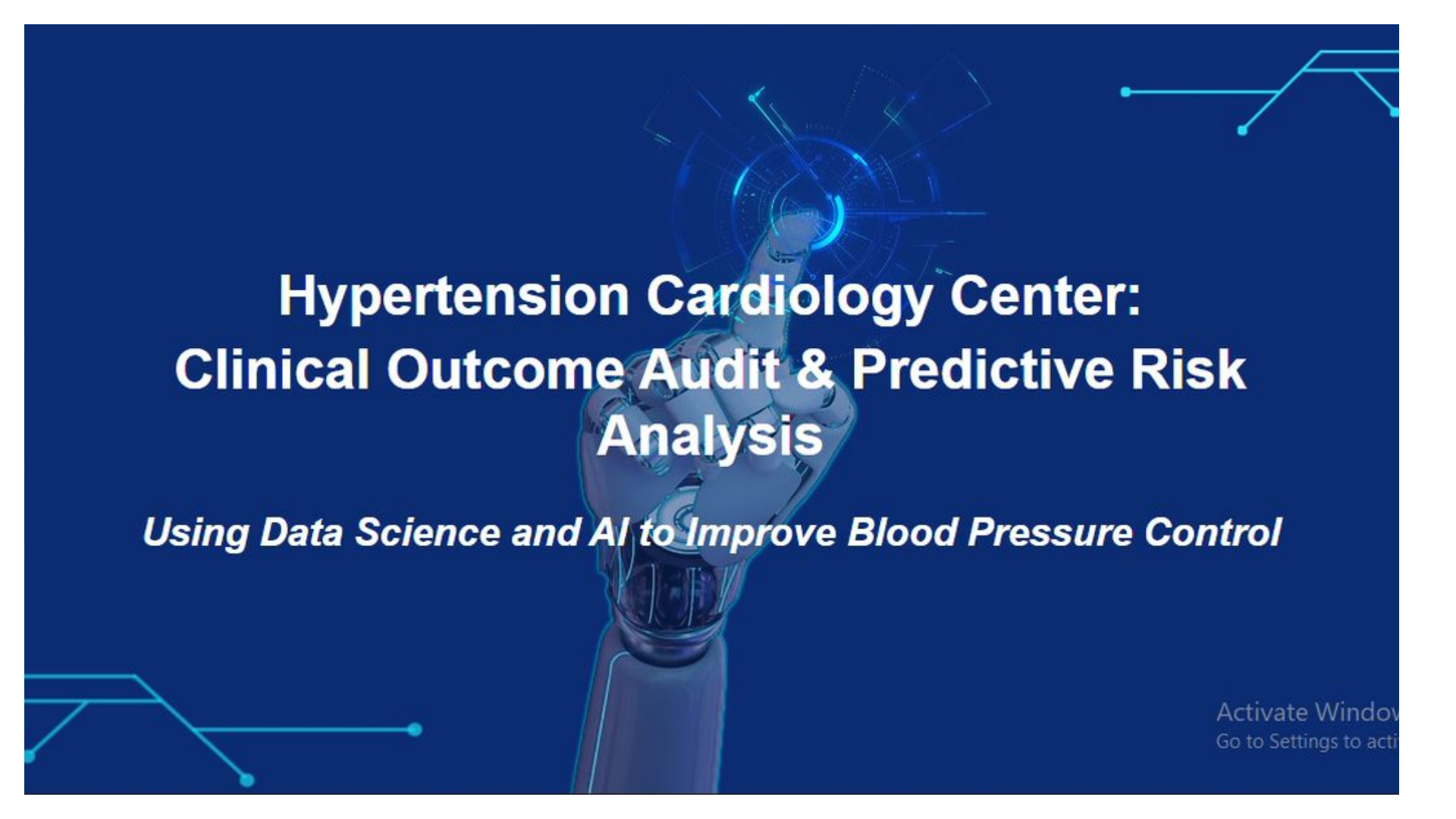




Hypertension Cardiology Center: Clinical Outcome Audit & Predictive Risk Analysis

Using Data Science and AI to Improve Blood Pressure Control

Presentation Structure

1. **Introduction**
2. Methodology
3. Predictive Modeling
4. Clinical & Behavioral Insights
5. Risk Stratification
6. Recommendation
7. Limitation
8. Conclusion



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Introduction



Objective of the Study

Analysis of 350 hypertensive patients

Treatment Optimization Goals

Identify better treatment strategies based on patient profiles



Key Insights from Data Analysis

Found a 91% failure rate in BP control at 3 months

Healthcare Solution

Built a model to identify high-risk patients early



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Clinical Context & Problem Statement

Uncontrolled hypertension increases the risk of serious cardiovascular conditions.

Early identification of high-risk patients is critical.

This Project Answers

- *What predicts BP control failure at 3 months?*
- *Which drug works best for different patient conditions?*

BP Control Failure = BP Controlled (No)

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Approach & Tools

Data cleaning and preparation

Feature creation from clinical variables

Machine learning model built in Python

Insights and visuals created in Power BI

DATA



ANALYSIS



MODEL

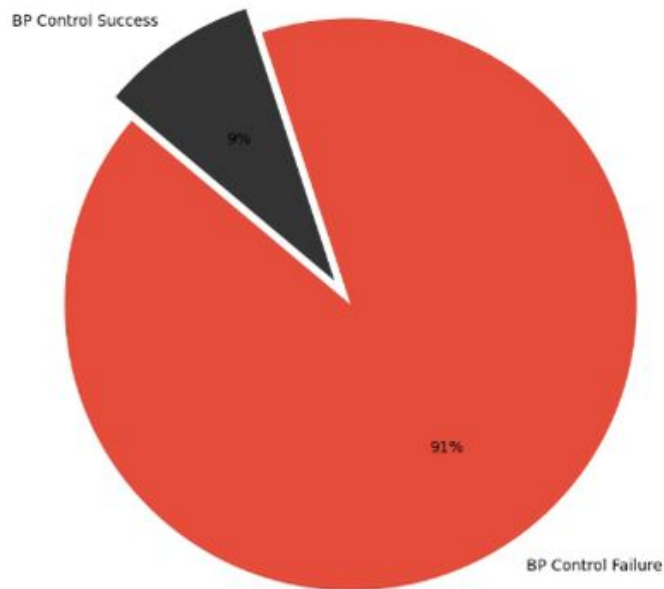


INSIGHTS

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Data Preparation & Class Imbalance

Training Data: Before SMOTE
(Imbalanced Reality)

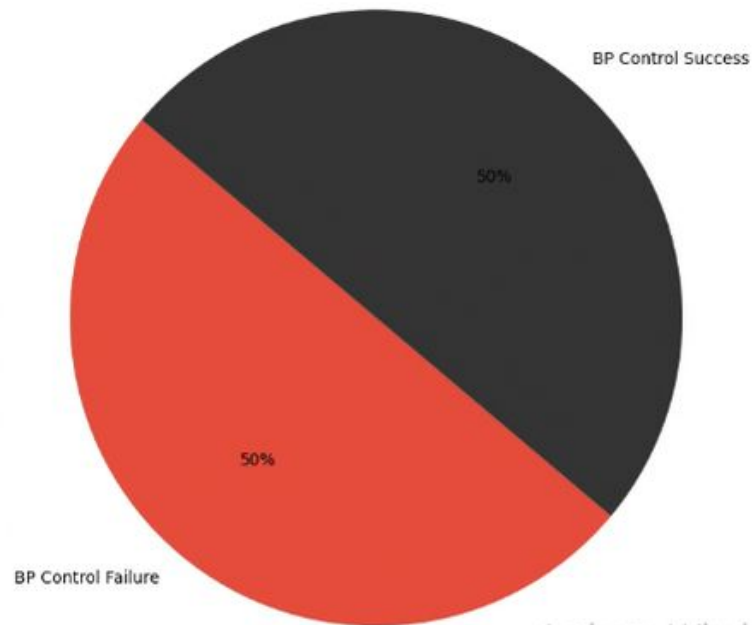


- **Dataset had 91% Failure Cases**
- **This imbalance can bias the model**
- **Applied SMOTE to balance the data**
- **This improved the model's ability to learn success patterns**

SMOTE Application for Balance

**Utilized SMOTE
to
generate synthetic
samples, effectively
addressing the class
imbalance issue.**

Training Data: After SMOTE
(Balanced for Model Training)

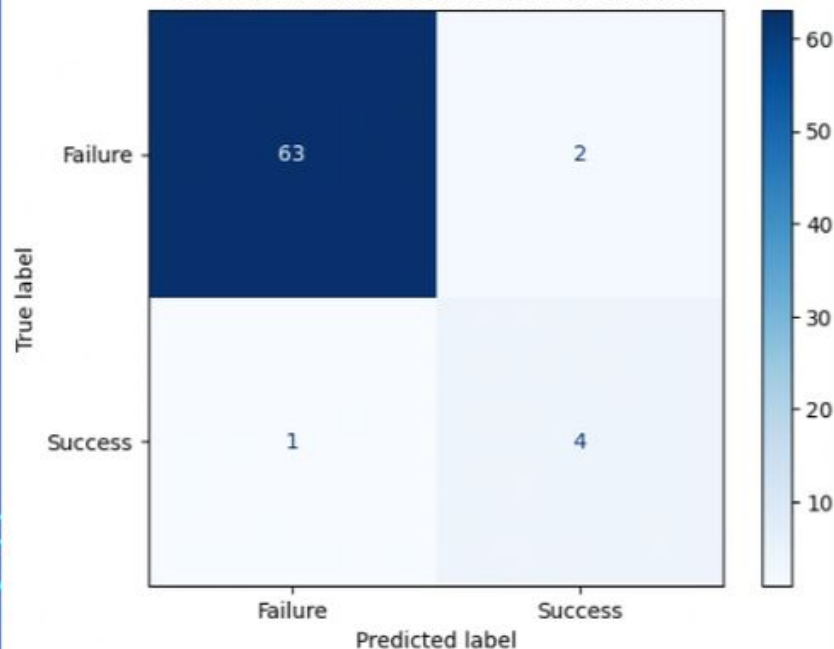


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Predictive Model Performance

Model used: Random Forest

Confusion Matrix: Real-World Performance

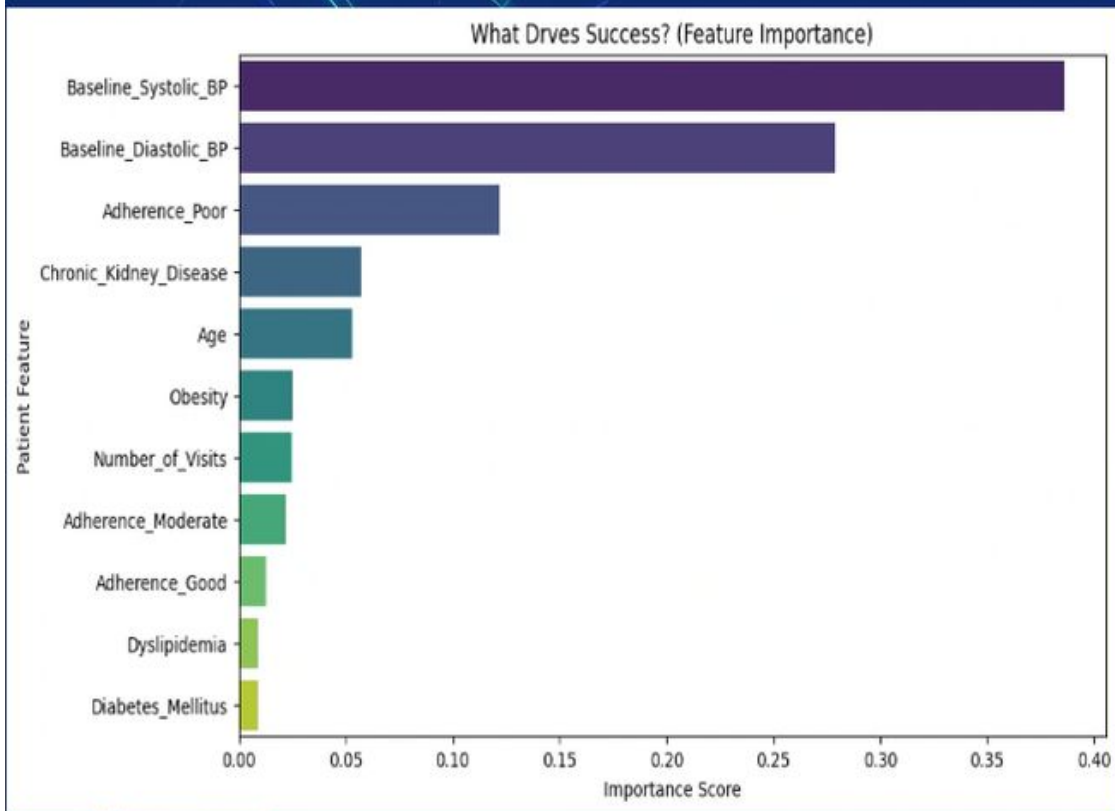


Designed to identify high-risk patients early

Strong performance in detecting likely failures

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Key Drivers of BP Control



Both medical condition and patient behavior affect outcomes

Top factors affecting BP control



Baseline Systolic Blood Pressure

Medication Adherence (Poor)

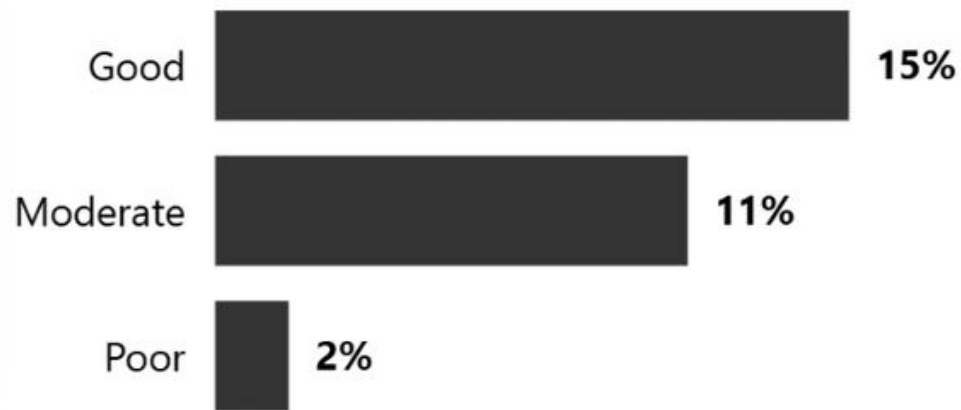


Chronic Kidney Disease

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Medication Adherence Insight

Success Rate by Medication Adherence



● Strong link between adherence and success

● Good adherence → higher BP control

● Poor adherence → high failure rate

Improving adherence can significantly improve outcomes

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Drug Class Effectiveness

Drug Class Efficacy by Outcome

Antihypertensive_Class	Success Rate %
Combination Therapy	21%
Thiazide Diuretic	13%
Beta Blocker	10%
ARB	6%
Calcium Channel Blocker	5%
ACE Inhibitor	

Top performing treatments

- Combination therapy
- Thiazide diuretics

Important Finding

ACE inhibitors showed no observed success in this dataset

This suggests a possible mismatch between drug choice and patient profile

Comorbidity-Based Treatment Insights

Treatment effectiveness depends
on patient condition

→ CKD patients → respond better to
Thiazide Diuretics

Diabetic patients → respond
better to Combination Therapy



→ Complex cases → need
targeted treatment

A one-size-fits-all approach does not work

Drug Class Efficacy by Outcome

Antihypertensive_Class	Success Rate %
Combination Therapy	25%
Thiazide Diuretic	11%
Beta Blocker	8%
ARB	6%
Calcium Channel Blocker	4%
ACE Inhibitor	

Drug Class Efficacy by Outcome

Antihypertensive_Class	Success Rate %
Thiazide Diuretic	18%
Combination Therapy	12%
Beta Blocker	6%
Calcium Channel Blocker	3%
ARB	3%
ACE Inhibitor	

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High-Risk Patient Identification

High-risk patients typically have

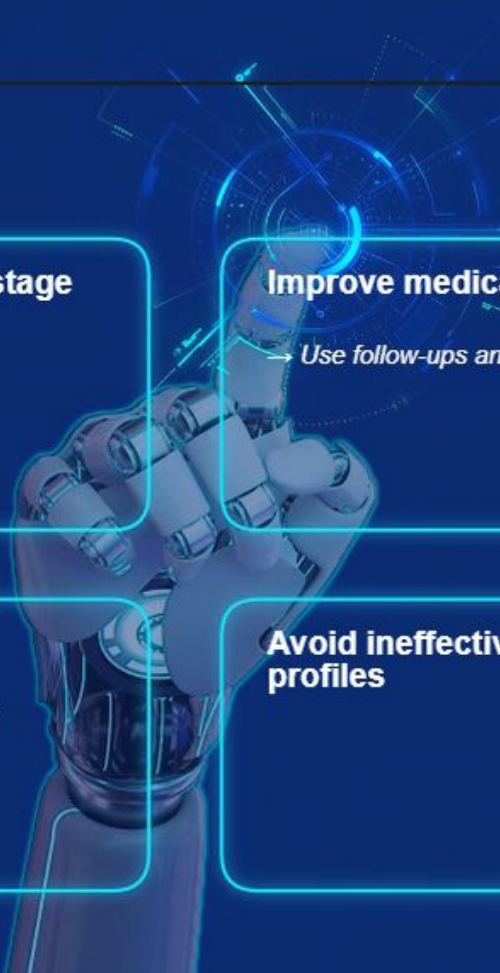
- High baseline BP
- Poor adherence
- Multiple health conditions

These patients can be identified early using the model

Baseline BP vs. Age: Identifying High-Risk Profiles



Clinical Recommendations



Use predictive model at prescription stage

→ *Identify high-risk patients early*

Improve medication adherence

→ *Use follow-ups and patient monitoring*

Optimize treatment choices

→ *Prefer combination therapy and thiazide diuretics*

Avoid ineffective treatments for certain profiles

Study Limitations



Dataset limited to 350 patients



Observational data, not experimental

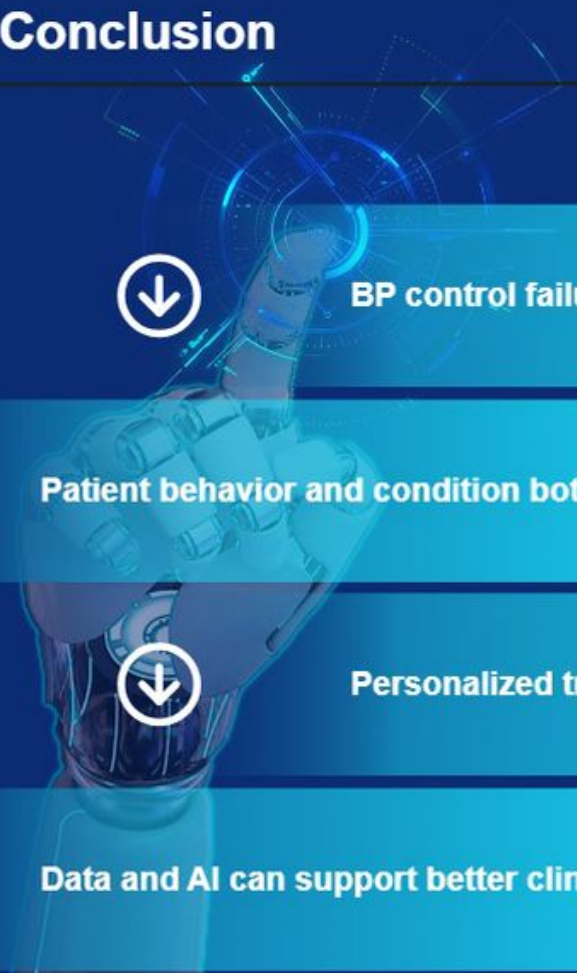


Results may not apply to all populations



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Conclusion



BP control failure can be predicted using data

Patient behavior and condition both matter



Personalized treatment improves outcomes

Data and AI can support better clinical decision-making



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Clinical Outcome Audit & Risk Analysis

Gender

Female

Male

Chronic_Kidney_Disease

No

Yes

Diabetes_Mellitus

No

Yes

Dyslipidemia

No

Yes

Obesity

No

Yes

Total Patient

350

Success Rate

9%

Avg Sys BP (Baseline)

166

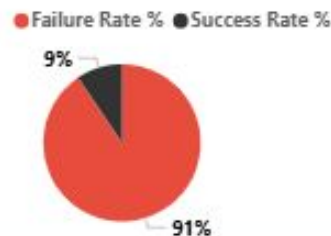
Avg Dia BP (Baseline)

105

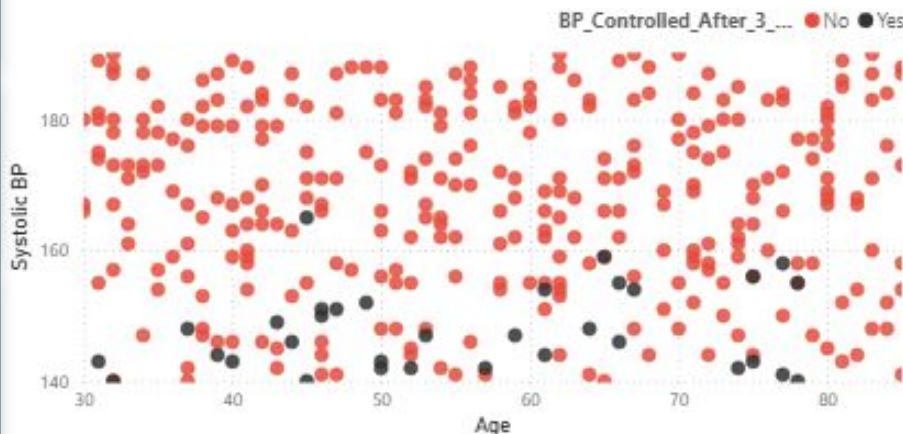
Success Rate by Medication Adherence



Success vs. Failure



Baseline BP vs. Age: Identifying High-Risk Profiles



Key influencers Top segments

What influences

BP_Controlled_After_3_Months to be

Yes

?

When...

...the likelihood of BP_Controlled_After_3_Months being Yes increases by

Average of Age is 78 or less

Infinity

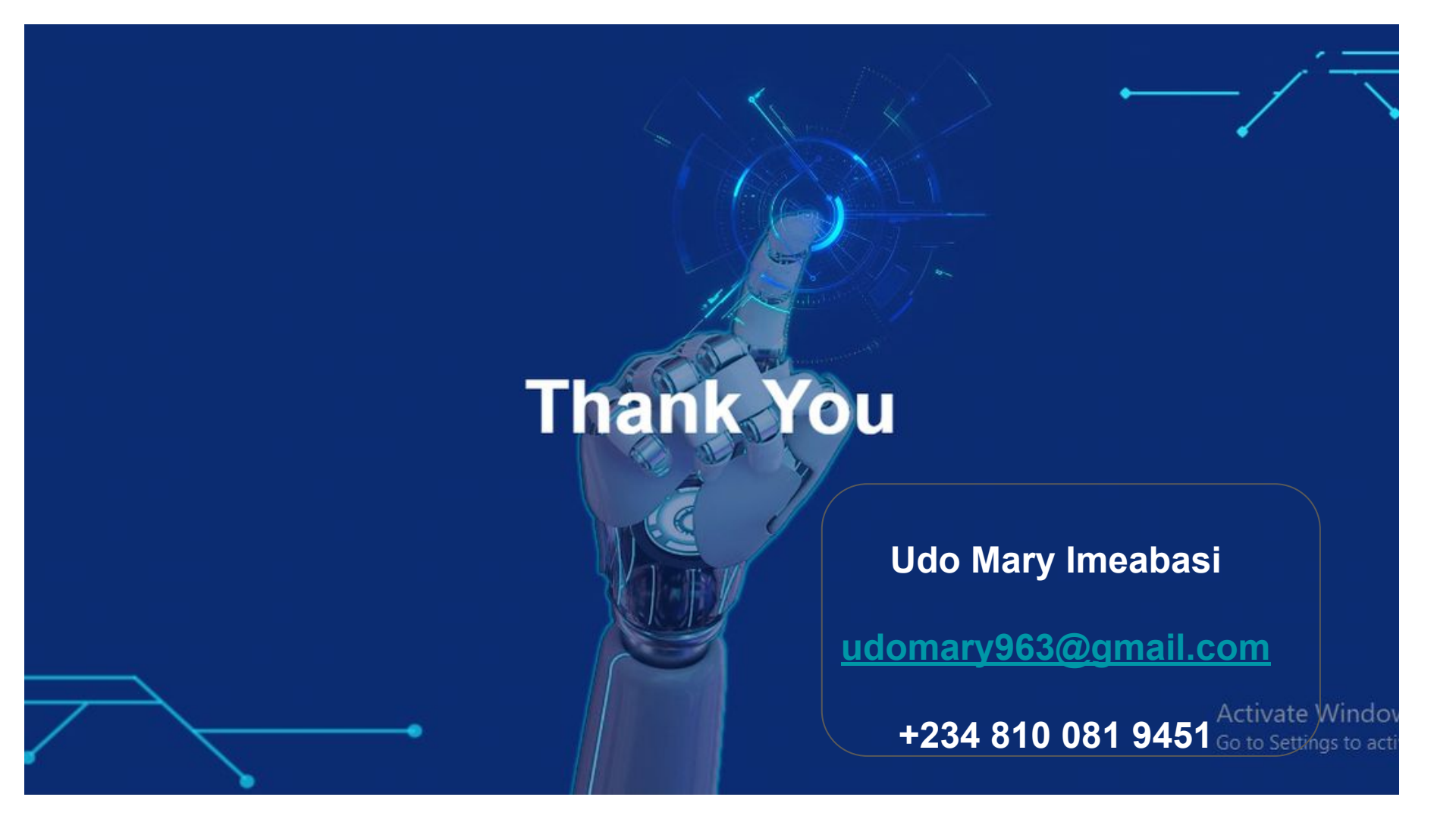
Antihypertensive_Class is Combination Therapy

2.94x

Drug Class Efficacy by Outcome

Antihypertensive Class	Success Rate %
Combination Therapy	21%
Thiazide Diuretic	13%
Beta Blocker	10%
ARB	6%
Calcium Channel Blocker	5%
ACE Inhibitor	5%

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Thank You

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